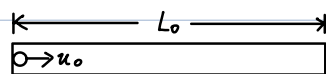
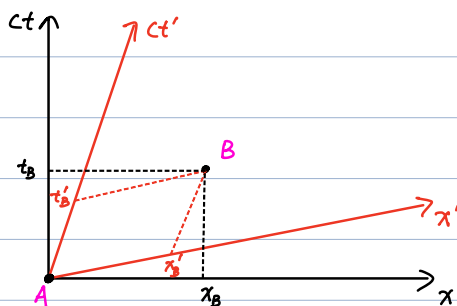


Example.

K' frame (spaceship moving at speed v)



Q: in K frame (earth), time interval of ball reaching other end?



A: ball at left

B: ball hits right

$$\begin{cases} x'_B = L_0 \\ t'_B = \frac{L_0}{u_0} \end{cases}$$

$$\Rightarrow \begin{pmatrix} ct_B \\ x_B \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} c \frac{L_0}{u_0} \\ L_0 \end{pmatrix} = \begin{pmatrix} \gamma c \frac{L_0}{u_0} + \beta\gamma L_0 \\ \beta\gamma c \frac{L_0}{u_0} + \gamma L_0 \end{pmatrix}$$

$$\Delta t = t_B - t_A$$

$$= \gamma \frac{L_0}{u_0} + \beta\gamma \frac{L_0}{c}$$

$$\begin{aligned} u &= \frac{x_B - x_A}{t_B - t_A} \\ &= \frac{\beta\gamma c \frac{L_0}{u_0} + \gamma L_0}{\gamma \frac{L_0}{u_0} + \beta\gamma L_0/c} \\ &= \frac{u_0 + v}{1 + \frac{u_0 v}{c^2}} \end{aligned}$$

Note: if blindly use "time dilation":

$$\Delta t = \gamma \frac{L_0}{u_0} \quad \text{X}$$

"with respect to"

why? "time dilation" for clock that is stationary wrt K'

General Relativity - Introduction

- Inertial frame $\rightarrow$ Non-inertial frame
- Gravity $\rightarrow$ Curved space & time (Riemann Geometry)

- Gravitational mass (m_g) 引力质量

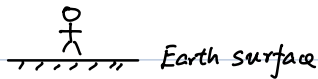
$$F = m_g g$$

- Inertial mass (m_i) 惯性质量

$$F = m_i a$$

$$m_g = m_i \equiv m \Rightarrow g \sim a ?$$

e.g.

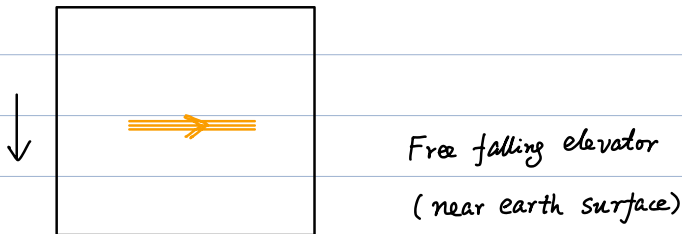


Both feel a force $f = mg$

Principle of Equivalence

At any space-time point, an arbitrary gravitational field $\Leftrightarrow$ an accelerated frame.

- Gravitational deflection of light



Elevator frame: light propagates in a straight line.

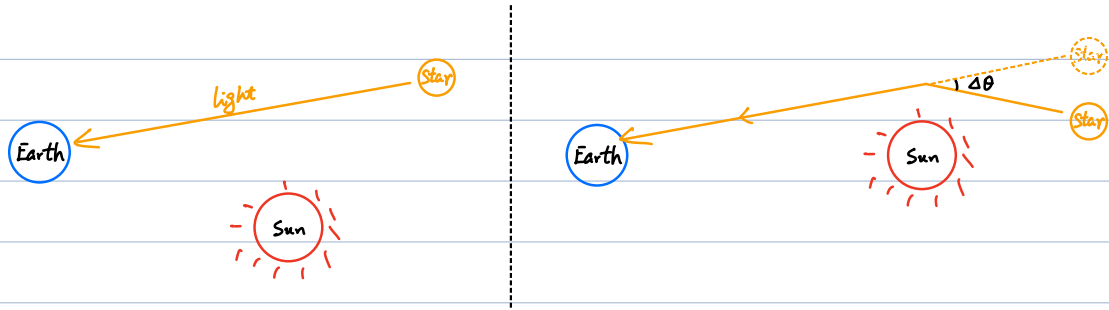
(Gravitational force & non-inertial force cancel)

Earth frame: light bends downwards

(due to gravity)



Experimental verification:



General relativity predicts:

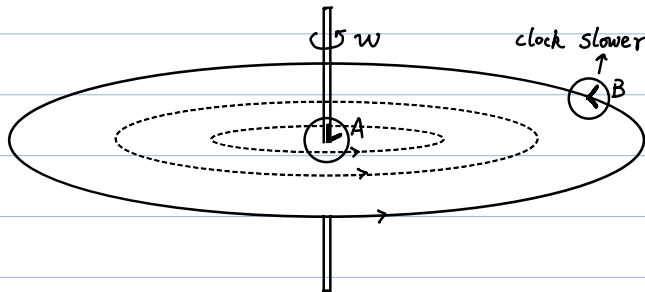
$$\Delta\theta = \frac{4GM_0}{R_0} \begin{matrix} \rightarrow \text{mass of sun} \\ \rightarrow \text{radius of sun} \end{matrix}$$

$$\approx 1.75''$$

Note: $\begin{cases} 1^\circ = 60' \\ 1' = 60'' \end{cases}$

Confirmed by Arthur Eddington & collaborators in 1919.

• Gravitational time dilation



In the rotating frame,

$$F_{\text{non-inert}} = mr\omega^2$$

$\Rightarrow$ Gravitational potential:

$$\varphi = -\int_0^r r'\omega^2 dr' = -\frac{1}{2}\omega^2 r^2 = -\frac{1}{2}v^2$$

Reminder: moving clock slows down

$$\Delta t = \gamma \Delta t_0$$

A ← → B

$$\Rightarrow \Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}}$$

$$= \frac{\Delta t_0}{\sqrt{1 + 2\varphi/c^2}}$$

Note: $\varphi < 0$

$$\Delta t > \Delta t_0$$

$\Rightarrow$ clock in gravitational field slows down.

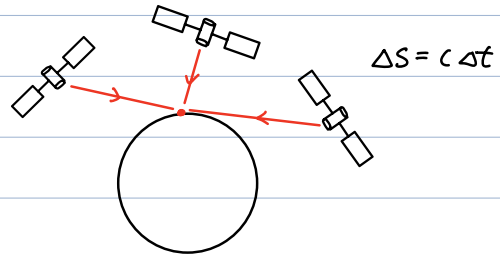
(3)

- GPS (Global positioning system, USA)

- 北斗

- GLONASS (Russia)

- Galileo (EU)



Key: determine Δt (clock on satellite)

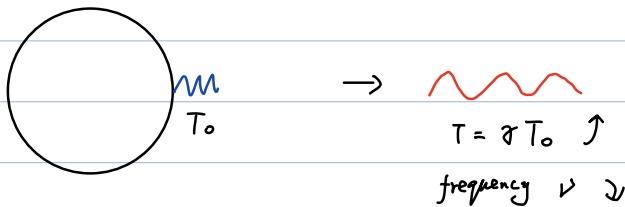
- special relativity: clock on satellite slowed by 7 $\mu\text{s/day}$

(satellite moving relative to Earth)

- general relativity: clock on satellite boosted by 46 $\mu\text{s/day}$

(gravity at satellite weaker)

- Gravitational red shift



Another way to understand:

photon overcomes gravitation to escape

$$\Rightarrow E = h\nu \downarrow \Rightarrow \nu \downarrow$$

Note: ν (Greek letter for frequency)

Not velocity v)

- Spacetime curvature

spacetime curved due to presence of massive objects

evidences:

- light travels along "null geodesics" (shortest possible path)

- 近日点
perihelion precession of Mercury

- Gravitational waves (LIGO 2016, Nobel Prize 2017)